FoodShare – Product Requirements Document (PRD)

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Overview

FoodShare is a hyperlocal web platform that moves surplus edible food from donors to nearby receivers within hours, using volunteers for the last mile.

Problem. Restaurants, grocery stores and households throw away edible food every day, while shelters, community fridges and low-income families nearby go without. The missing piece is real-time coordination: nobody knows what is available, where it is, or how long it stays safe.

Solution. A three-sided marketplace. Donors post surplus, receivers (NGOs, shelters) claim it, and volunteers deliver it. Every party sees live status, and listings appear only inside a strict radius (default 5 km) so pickups stay fast and food stays fresh.

Goals and success metrics

The product succeeds if surplus food is claimed and delivered before it expires. The targets below are proposed starting points to be tuned after the pilot.

Goal	Metric	Proposed target
Rescue food before it spoils	Share of listings reaching Delivered	70% or more
Fast coordination	Median time from Available to Claimed	Under 20 minutes
Fast delivery	Median time from Claimed to Delivered	Under 90 minutes
Supply growth	Active donors per month in pilot area	50 or more
Volunteer capacity	Share of claimed listings picked up within 30 minutes	60% or more
Visible impact	Meals saved and CO2 avoided shown on dashboard	Live, updated per delivery

Meals saved = delivered kg divided by kg per meal; CO2 avoided = delivered kg times an emission factor. Both factors are configurable (defaults 0.4 kg per meal, 2.5 kg CO2e per kg).

Users and personas

Three roles, each with its own permissions and home screen.

Role	Who they are	What they want	Main actions
Donor	Restaurants, grocers, caterers, households	Give away surplus quickly with little effort	Post a listing, set pickup window, cancel, view their impact
Receiver	NGOs, shelters, community fridges	Find usable food nearby and know when it arrives	Browse nearby listings, claim, track delivery
Volunteer	Individuals with a bike or car	Short, clear delivery tasks near them	See claimed pickups, accept pickup, mark delivered

Example: Priya runs a bakery and has 6 kg of unsold bread at 9 pm. She posts it in under a minute, a nearby shelter claims it, and a volunteer delivers it before it spoils.

Functional requirements

Priority uses MoSCoW: Must, Should, Could.

ID	Requirement	Acceptance criteria	Priority
FR-1	Register and log in with email and password, choosing a role	Role is stored and cannot be changed by the user	Must
FR-2	Role-based access control	Donors cannot claim, receivers cannot deliver, volunteers cannot post; wrong role returns 403	Must
FR-3	Donor creates a listing	Title, category, quantity (kg), expiry, address and map location required	Must

ID	Requirement	Acceptance criteria	Priority
FR-4	Nearby listings by radius	Only Available, unexpired listings within the chosen radius (default 5 km), nearest first	Must
FR-5	Receiver claims a listing	Only one receiver can win; a second claim gets a clear conflict message	Must
FR-6	Volunteer accepts a claimed pickup	Listing moves to In Transit and shows the volunteer	Must
FR-7	Volunteer marks delivered	Only the assigned volunteer can do it; listing moves to Delivered	Must
FR-8	Live status updates	Status changes reach all open clients within 2 seconds without refresh	Must
FR-9	Donor cancels an unclaimed listing	Allowed only while Available	Should
FR-10	Impact dashboard	Total meals saved, CO2 avoided, kg by category, daily trend, top donors and receivers	Must
FR-11	My activity list per role	Donors see posts, receivers see claims, volunteers see deliveries	Should
FR-12	Auto-expire listings	Past-expiry listings leave the nearby feed	Should
FR-13	Notifications (in-app)	Donor told on claim and delivery; volunteers told of new claimed pickups nearby	Could
FR-14	Ratings and trust badges	Receivers and donors rate each handover	Could

Non-functional requirements, scope and risks

Non-functional

- Performance: nearby query returns in under 300 ms for 100,000 listings.

- Real-time: status events delivered in under 2 seconds on a normal connection.
- Security: passwords hashed, JWT sessions, role checks on every write endpoint.
- Availability: single-region deployment with daily database backups.
- Usability: works on mobile browsers; core flows reachable in three taps.

Out of scope for v1

- Payments or monetary donations.
- Route optimisation and live GPS tracking of volunteers.
- Native mobile apps.
- Automated food-safety verification.

Risks and open questions

- Food safety and liability: needs a donor declaration and a clear disclaimer; legal review required before launch.
- Cold start: too few volunteers makes claimed food sit; seed volunteers through partner NGOs first.
- Fake or spam listings: consider donor verification for organisations.
- Open question: should receivers be approved by an admin before they can claim?
- Open question: which city is the pilot area, and what radius fits it?